

HIGH-RESISTANCE METER



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This resistance meter can measure resistances up to 200 mega-ohms. The value of the resistance is indicated on a linear meter scale (0-100 μ A). The operation of the circuit is based on sensing the voltage drop across a reference resistor. The voltage sensing IC used here is a CMOS NAND gate (CD4011) whose high impedance makes it suitable for sensing the voltage drop across high resistances.

Connect the high resistance (R_x) to be measured in series with the reference resistor (here $R_{ref} = 10$ mega-ohms). Apply a variable high voltage (up to 100V) at the other end of the unknown resistance (R_x) such that the voltage drop across the reference resistor (R_{ref}) is equal to the threshold voltage (V_{thr}) of gate N3 of IC1.

At this point, the voltage drop across the reference resistor (R_{ref}) is sensed by gate N3 and its output goes low. Gate N4 inverts the low input to light up LED1. Now stop varying the high voltage. The resistance value is indicated on the meter directly.

At this time, the expression for the unknown resistance R_x by applying Ohm's

law can be written as:

$$R_x = \frac{V}{V_{thr}} R_{ref} - R_{ref} \dots \dots \dots \text{Eq. (1)}$$

where 'V' is the voltage applied across the R_x - R_{ref} series combination (i.e., the voltage at the emitter of transistor T4) and V_{thr} is the threshold voltage of gate N3, which is also the voltage drop across the reference resistor (R_{ref}) when LED1 glows.

From Eq. (1), it is clear that the second term ($-R_{ref}$) has to disappear in order to have a direct linear relationship between V and R_x . The equation can be rearranged as:

$$V = \frac{R_x}{R_{ref}} V_{thr} + V_{thr} \dots \dots \dots \text{Eq. (2)}$$

This shows that by subtracting V_{thr} from V, the relationship between $V - V_{thr}$ and R_x becomes directly linear (without any offset). This subtraction is achieved by using transistor BC557 (T5) as the voltage follower with potmeter VR6. Adjusting the potmeter to get V_{thr} at the emitter of BC557 produces the required offset voltage at the other end of the meter to nullify the second term (V_{thr}) such that the current through the meter is proportional to R_x .

The high-voltage section consists of a high-voltage generator, which is basically an oscillator driven by step-down transformer X1. A 230V AC primary to 6V-0-6V AC, 150mA secondary transformer is used here. To derive the power supply, the transformer output is rectified by a bridge rectifier comprising diodes D3 through D6, filtered by capacitor C2 and regulated by zener diodes ZD1, ZD2 and ZD3 (25V, 50V, 100V) in combination with transistor BD115 (T3). Diodes D7 and D8 (each 1N4148) provide the required additional drop in voltage across the base-emitter junction of transistors T3 and T4 (each BD115).

